PD Submission Doc

Name: Grant Li
Discord: gran07

Part 1

1.1a

Number of instances: 6159 cell instances

Total cell area: 328284.026438

1.1b

chip_top.gate.v lists gate netlist and SRAM macros that are produced from synthesis of the RTL. The purpose is to specify the cells that are used and how they are connected in the physical layout of the chip.

Part 2

2.1a

- Total cell area = 328284.026438, SRAM macro area = 235293.578125
- Non macro cell area = 328284.026438 - 235293.578125 = 92990.44831
- SRAM macro with halo = 2*(387.870+42)*(306.775+42) = 299855.8185
- Total area = 92990.44831 + 299855.8185 = 392846.2668
- Core area with 50% core utilization: (2)(392846.2668) = **785692.5336**
    - X_dim = 2*Y_dim = **1253.549**
    - Y_dim = **626.774**
    - x_Dim*Y_dim = 2*Y_dim*Y_dim = 785692.5336; Y_dim = 626.774

2.2a

```
floorPlan -site CoreSite -adjustToSite \
        -s $x_dim $y_dim \
        $left_offset $bottom_offset $right_offset $top_offset
```

2.2b

placeInstance $macro_name_1 248.5045 169.5995 -fixed
placeInstance $macro_name_2 636.3745 169.5995 -fixed

2.3a

- Helps ensure there is stable voltage around the macro
- Macros are larger and may require greater power consumption so providing them with power rings can help with even distribution of power to internals
- Prevent IR drop due to resistance in one area of the chip
    - Less resistance due to larger wires

2.3c

```
addRing \
    -around cluster \
    -jog_distance 0.4 \
    -nets {VSS VDD} \
    -type block_rings \
    -layer {bottom met5 top met5 right met4 left met4} \
    -width 6.2 \
    -spacing 3.2 \
    -offset 5
```

2.4a

checkFPlan -outFile util.txt -reportUtil

Checking Preroutes.....

No. of regular pre-routes not on tracks : 0

Reporting Utilizations.....

Core utilization = 50.001009

TU for group group_SRAMs = 100.000000

Effective Utilizations

Average macro density = 0.000.

Average module density = 0.188.

Pure std cell Density 0.187862

 Density for the design = 0.188.

= stdcell_area 48893 sites (93112 um^2) / alloc_area 260260 sites (495639 um^2).

Pin Density = 0.05896.

= total # of pins 24260 / total area 411475

Part 3

3.1a

Startpoint: cpu/saved_instr_reg_20_/Q
Endpoint: cpu/reg_inst/registers_reg_15__10_/D
Path delay: 7.819 ns
Required time: 7.462 ns
Slack: -0.358 ns

After using –max_paths, my worst paths begin from my saved_instr register through the execute logic and ending at a register. This common behavior suggests a path bottleneck rather than from a single cell type, maybe the logic depth is slightly too high for cpu execution.

Also the paths travel through a MUX cell which has a higher delay than others (0.500 ns)

3.2a

```
proc report_tns {} {
    set paths [report_timing -collection -max_paths 1000000]
    set tns 0.0

    foreach_in_collection path $paths {
        set path_slack [get_db $path .slack]


        if {$path_slack < 0.0} {
            set tns [expr {$tns + $path_slack}]
        }
    }
    return $tns
}
```

3.2b

WNS: -0.358 ns

TNS: -131.81758843

Part 4

4.1a

Added flag –gate_clock to compile_ultra in both branches (with or without DFT)

- WITH_DC_ULTRA is set as 1 in config so clock gates should be added

4.2a

All prefixed by ICG

- ICGPRECX1
- ICGPRECX2
- ICGPRECX4
- ICGX1
- ICGX2
- ICGX4

4.2b

ICGPRECX1 and it is instantiated 34 times

4.3a

Needed to use –unconstrained because analyzing timing through clock gate doesn't involve going from FF to FF so not constrained by timing (like setup time)

```
#####################################################################
#  Generated by:      Cadence Innovus 25.11-s102_1
#  OS:                Linux x86_64(Host ID ece-rhelsrv)
#  Generated on:      Fri Oct 16 12:05:12 2026
#  Design:            chip_top
#  Command:           report_timing -through cpu/u_fetch/clk_gate_pc_o_reg/latch -unconstrained
#####################################################################
Path 1:Endpoint:   cpu/u_fetch/pc_o_reg_11_/CK (^) (unconstrained output)
Beginpoint: cpu/state_reg_1_/Q          (v) triggered by trailing edge of
'coreClk'
Arrival Time                  2.449
Analysis View: typical_analysis_view
     + Clock Network Latency (Prop)  0.054
     = Beginpoint Arrival Time       0.054
     +-------------------------------------------------------------------------------+
     |              Instance               |     Arc      |   Cell    | Delay | Arrival |
     |                                     |              |           |       |  Time   |
     |-------------------------------------+--------------+-----------+-------+---------|
     | cpu/state_reg_1_                    | CK ^         |           |       |   0.054 |
     | cpu/state_reg_1_                    | CK ^ -> Q v  | DFFX1     | 0.369 |   0.423 |
     | cpu/U11                             | A v -> Y ^   | NOR2X1    | 0.303 |   0.726 |
     | cpu/U12                             | A ^ -> Y v   | NAND3X4   | 0.226 |   0.952 |
     | cpu/u_fetch/U6                      | B v -> Y ^   | NOR3X1    | 0.810 |   1.761 |
     | cpu/u_fetch/U140                    | B ^ -> Y ^   | OR2X1     | 0.207 |   1.969 |
     | cpu/u_fetch/clk_gate_pc_o_reg/latch | E ^ -> ECK ^ | ICGPRECX1 | 0.336 |   2.304 |
     | cpu/u_fetch/CTS_ccl_a_buf_00140     | A ^ -> Y ^   | CLKBUFX4  | 0.144 |   2.448 |
     | cpu/u_fetch/pc_o_reg_11_            | CK ^         | DFFX1     | 0.001 |   2.449 |
     +-------------------------------------------------------------------------------+
```

4.3b

WNS = -5.271 ns

TNS = -1870.87769224 ns

Part 5

5a

Faster cells draw more current but are larger which takes up more area. Also using faster switching cells may draw more static/leakage power.

For this project, we are mainly focused on closing setup timing so increasing the area or drawing more static power is not extremely important. This tradeoff might be significant for a smaller chip

5b

Corners are extreme limits based on variations in semiconductor manufacturing that can affect transistor performance.

SS: Both PMOS and NMOS transistors have slower switching. May cause signals to arrive late (setup time violation)

FF: Both PMOS and NMOS transistors have fast switching. May cause signals to arrive too early (hold time violation)

TT: Typical switching speed for PMOS and NMOS transistors. Used to measure the typical timing of the logic

5.1a

Setup the paths to HS cell libraries for FF, TT, SS. Added these new libs for target and link library variable

```
# HS best case library
set ::LIB_HS_FF_NAME  "HS_sky130_ff_1.98_0"
set ::LIB_HS_FF_DB    "$::LIB_DBS_PATH/$::LIB_HS_FF_NAME.db"

# HS typical case library
set ::LIB_HS_TT_NAME  "HS_sky130_tt_1.8_25"
set ::LIB_HS_TT_DB    "$::LIB_DBS_PATH/$::LIB_HS_TT_NAME.db"

# HS worst case library
set ::LIB_HS_SS_NAME  "HS_sky130_ss_1.62_125"
set ::LIB_HS_SS_DB    "$::LIB_DBS_PATH/$::LIB_HS_SS_NAME.db"
```

```
set ::target_library "$::LIB_TT_DB $::LIB_HS_TT_DB"
set ::link_library   "$::LIB_TT_DB $::LIB_SS_DB $::LIB_FF_DB $::LIB_HS_TT_DB $::LIB_HS_SS_DB $::LIB_HS_FF_DB"
```

5.1b

```
NAND4X1 U137 ( .A(n56), .B(n55), .C(n54), .D(n53), .Y(n57) );
NOR2X1 U138 ( .A(rd_addr_i[4]), .B(rd_addr_i[3]), .Y(n1491) );
AOI21X1 U143 ( .A0(n1491), .A1(n62), .B0(rst_i), .Y(n63) );
CLKAND2X2 U144 ( .A(rd_write_en_i), .B(n63), .Y(n1490) );
CLKBUFX2 U145 ( .A(n1490), .Y(n1031) );
AND2X2_HS U146 ( .A(rd_data_i[30]), .B(n1031), .Y(n1516) );
CLKBUFX2 U147 ( .A(n228), .Y(n1341) );
CLKBUFX2 U148 ( .A(n229), .Y(n1340) );
AOI22X1 U149 ( .A0(reg_values_o[899]), .A1(n1341), .B0(n1340), .B1(
    reg_values_o[675]), .Y(n67) );
```

5.2a

.lib includes logic, timing, and power of cells

.lef (library exchange format) provides physical manufacturing details such as cell shapes/sizes, pins, routing which is needed by APR tool to correctly allocate space and route cells in the design

5.2c

WNS: -2.248 ns

TNS: -1434.842114 ns

```
Path 1: VIOLATED Clock Gating Setup Check with Pin cpu/reg_inst/clk_gate_
registers_reg_31_/latch/CK
Endpoint:   cpu/reg_inst/clk_gate_registers_reg_31_/latch/E (v) checked with
leading edge of 'coreClk'
Beginpoint: cpu/saved_instr_reg_8_/Q                        (v) triggered by
trailing edge of 'coreClk'
Path Groups: {clock_gating_default}
Analysis View: typical_analysis_view
Other End Arrival Time         -0.807
- Clock Gating Setup            0.300
+ Phase Shift                   5.000
= Required Time                 3.893
- Arrival Time                  6.142
= Slack Time                   -2.248
     Clock Rise Edge                2.500
     + Clock Network Latency (Prop) 0.168
     = Beginpoint Arrival Time      2.668
     +-----------------------------------------------------------------------------------------------+
     |                  Instance                  |     Arc     |    Cell    | Delay | Arrival | Required |
     |                                            |             |            |       |  Time   |   Time   |
     |--------------------------------------------+-------------+------------+-------+---------+----------|
     | cpu/saved_instr_reg_8_                     | CK ^        |            |       |   2.668 |    0.419 |
     | cpu/saved_instr_reg_8_                     | CK ^ -> Q v | DFFX1      | 0.355 |   3.023 |    0.775 |
     | cpu/reg_inst/FE_OFC491_rd_addr_1           | A v -> Y v  | BUFX2      | 0.409 |   3.432 |    1.184 |
     | cpu/reg_inst/U139                          | B v -> Y ^  | NOR3X1     | 1.218 |   4.650 |    2.402 |
     | cpu/reg_inst/U143                          | A1 ^ -> Y v | AOI21X1    | 0.587 |   5.237 |    2.989 |
     | cpu/reg_inst/U144_dup                      | B v -> Y v  | AND2X2_HS  | 0.273 |   5.510 |    3.262 |
     | cpu/reg_inst/U1198                         | C v -> Y ^  | NAND3X2_HS | 0.452 |   5.962 |    3.714 |
     | cpu/reg_inst/U1589                         | A1 ^ -> Y v | OAI21X1_HS | 0.178 |   6.140 |    3.892 |
     | cpu/reg_inst/clk_gate_registers_reg_31_/latch | E v      | ICGPRECX1  | 0.001 |   6.142 |    3.893 |
     +-----------------------------------------------------------------------------------------------+
```

5.2d

Slack improved from 4.3b after adding clock gating to the design. The WNS path is still the same as in 4.3b which involves going from saved_instr reg to the clock_gate’s enable signal.

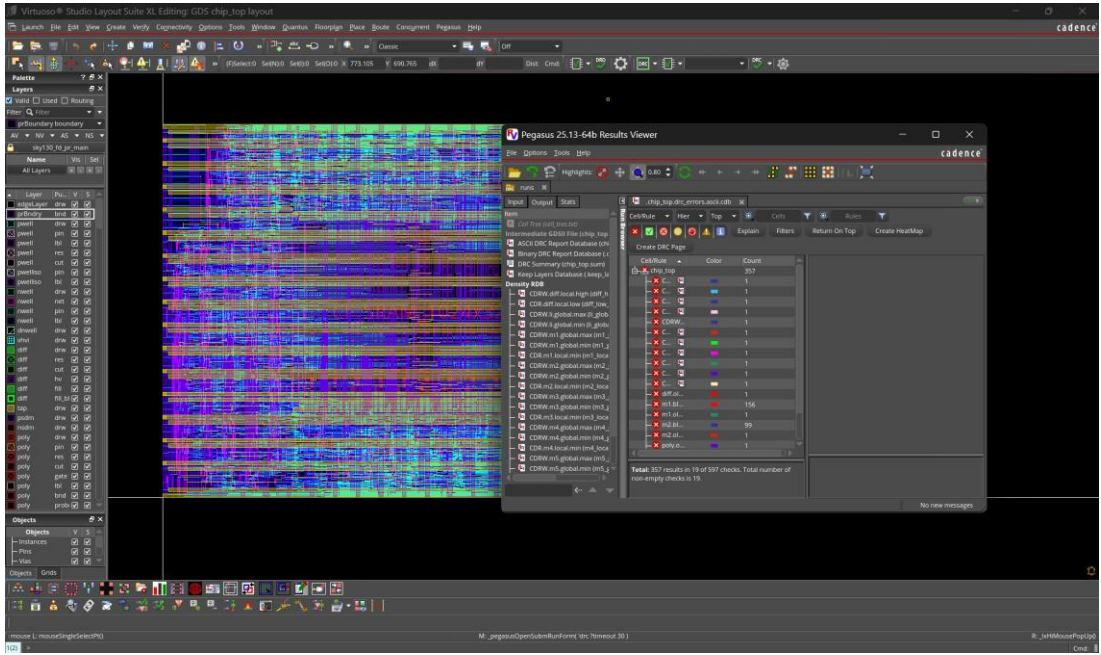
Compared to 3.2b, the slack is still worse as adding clock gates increased the negative slack by a lot possibly because clock gating introduces enable signals that need to be properly setup before the clock can be provided to certain cells.

Part 6

6.1

The main source of errors came from routing on a metal layer overlapping with areas where no wires should have been placed (ex. met1, met2). The errors could be power stripes overlapping the blockages, so maybe they could be rerouted with another layer in the same direction.

Another error is near the pins on the edges of the chip where the I/O shapes may have extended outside of the metal layer. The metal may need extending to cover the I/O pin polygons.



6.2

Most of my LVS errors are from shorts involving clock gates. It seems the errors relate to the clock gate enable pins being connected to a tie low net, so maybe an unintended connection is present here keeping the enable low. Looking at the netlist and comparing with the GDS layout would be useful to try and fix this.

